Muhammad Fayyaz

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~~Portfolio~~Portfolio Github

Technical Skills

Languages and DataBases: Python, Pandas, Numpy, SQL
Visualization Tools: Tableau, Power BI, Python, jupyter notebook
Other Skills: Data Science, Machine Learning Engineer, Business Analysis, Data Analysis, Customer Acquisition
Methods: *Supervised-Learning, Clustering, time Series, Un-supervised, Feature Engineering*

Experience/Projects

January 2024 – Jan 2026
Data Scientist
Punjab, Pakistan

1.Risk & Credit Default Predictions

- **Problem:** High rate of Non-Performing Loans (NPLs) due to manual or outdated credit assessment processes.
- **Solution:** Developed a **Machine Learning-based Credit Scoring Model** using **Logistic Regression** and **XGBoost** to evaluate the creditworthiness of loan applicants.

2. Customer Churn Analysis (Retention)

- **Problem:** Increasing number of customers closing their accounts or moving to competitors, leading to a loss in deposits.
- **Solution:** Built a **Customer Churn Model** using **Random Forest** to identify "at-risk" customers based on declining transaction volumes and account activity.
- *Developed a churn prediction engine that identified high-risk customers with **85% precision**, enabling the marketing team to execute retention campaigns that saved **$1.2M** in potential lost deposits.*

• 3. Fraud Detection in Digital Transactions

- **Problem:** Rising frequency of fraudulent credit card and mobile banking transactions.
- **Solution:** Engineered an **Anomaly Detection System** using **Isolation Forests** and **Neural Networks** to flag suspicious real-time transactions.
- *Automated fraud detection for digital banking, reducing manual investigation time by **40%** and preventing an estimated **PKR 50M** in annual fraudulent losses*

• 4. ATM Cash Optimization (Time Series)

- **Problem:** ATMs either running out of cash (bad customer experience) or holding too much idle cash (opportunity cost).
- **Solution:** Applied **Time Series Forecasting (ARIMA/Prophet)** to predict daily cash requirements for HBL's ATM network.
- *Optimized ATM cash replenishment schedules using time-series forecasting, reducing idle cash holdings by **15%** while maintaining a **99.9%** uptime for customers.*

Personal Projects/

Machine Learning Projects:

. I've worked on mutiples Machine Learning Models, like **Regression, Classification, Clustering, Ensembling & many more.**
. You may further visit on my github Repository & portfolio website.

Education

University of Education Lahore **Oct 2020 – May 2024**
Bachelor of Economics *Dera Ghazi Khan , Punjab*